Trust as hysteresis: analytically tractable strategies for the iterated prisoner's dilemma

Casey Crane

October 15, 2026

Abstract

Trust is fundamental to sustained cooperation in dyadic relationships, yet classical memory-one strategies for the iterated prisoner's dilemma (IPD) can only exhibit reciprocity, which is insufficient to model long-term dyadic trust. More sophisticated long-memory strategies exhibit richer behaviors, but are not analytically tractable, and can only be studied through simulation. We introduce a novel class of trust-based strategies for the IPD, built from elementary hysteretic components called hysterons. These four-hysteron (4H) strategies maintain an internal trust state that changes only when an accumulated memory of past interactions crosses a threshold. We develop a geometric framework where game histories are represented as points, thresholds as lines, and decisions as linear trajectories on the unit square, giving powerful tools for analyzing these strategies' long-term behaviors. Using this framework, we prove three main results with direct evolutionary interpretations. First, a deterrence theorem proves that trust is more expensive to rebuild than it is profitable to break. Second, we show that the optimal patterns of defections sustainable without breaking trust are exactly the lower Christoffel words, revealing a surprising connection to the combinatorics of words. Third, we analyze threshold selection in a noisy environment, where resilience to noise-induced defections must be balanced against vulnerability to exploitation. By providing an analytically tractable model of dyadic trust, this work offers a new lens for understanding the evolutionary mechanisms driving trust and cooperation between individuals. Because payoff differences translate directly to fitness differences, these results have direct implications for the evolution of cooperative strategies.

1 Introduction

The iterated prisoner's dilemma (IPD) explores the conflict between rational self-interest and collective cooperation, and is an important model for studying the evolution of cooperative behaviors among individually self-interested actors. From Axelrod's seminal tournament and the development of classical strategies [1, 2, 3, 4] to the discovery of zero-determinant strategies [5], this simple model continues to yield surprising, profound, and interpretable insights into the evolutionary incentives which shape cooperation, with deep connections to biological, sociological, and economic systems [6, 7, 8, 9].

Trust is a concept deeply intertwined with that of cooperation. It is commonly defined as "a psychological state comprising the intention to accept vulnerability based upon positive expectations of the intentions or behavior of another" [10, 11]. While trust has been studied extensively in iterated games, from the trust game [12] (and iterated versions [13]) to modern work on trust dynamics [14], its role in the IPD has primarily been explored through the lens of public reputation, partner selection [15, 16, 17], and indirect reciprocity [18, 19, 20]. These models, however, capture a fundamentally different phenomenon from the private, dyadic trust that develops between

two specific individuals through repeated interaction. The analytical study of dyadic trust in the IPD—trust as a persistent, history-dependent internal state rather than a public score—remains a significant gap in the literature.

One reason for this is that modeling dyadic trust requires both long memory and an internal state. Strategies with these properties—such as artificial neural networks (ANNs) or finite-state machines (FSMs)—are difficult to study analytically, and are therefore typically studied through simulation rather than through rigorous analysis [21, 22].

In this paper, we introduce a family of trust-based strategies for the IPD with memory, internal states, and threshold-based behavior, and that are analytically tractable. We realize these properties through the hysteron, an elementary model of hysteresis, and analyze a strategy which makes decisions based on the output of four hysterons with independent thresholds and states. The key technical innovation allowing this analysis is a geometrical framework—the memory square—in which infinite game histories are represented as points, trust thresholds are lines, and round outcomes are affine contractions. This framework allows us to prove rigorous results about the long-term behavior of these strategies, including the asymmetric cost of breaking trust, their vulnerability to patterned exploitation, and their resilience to noise. While we do not perform any evolutionary simulations in this paper, these analytical results have direct evolutionary interpretations, since payoff differences translate to fitness differences.

1.1 Trust-based strategies

While it is beyond the scope of this paper to give a comprehensive definition of trust—either behaviorally or mathematically—we would still like to clarify what we mean by a "trust-based strategy" for the IPD in terms of a few basic properties:

(T1) Internal state and history dependence: A trust-based strategy maintains an internal state which is updated on each round in response to the round outcome, and so implicitly depends on the full game history. It makes decisions based on this internal state, and possibly on the most recent round outcome.

(T2) Path dependence: The internal trust state changes only when the input crosses a threshold. Between these thresholds, the state is constant. Consequently, the current value of the input is not sufficient to determine the state; two histories with the same current input may yield different states.

(T3) Asymmetry: The conditions for losing trust are weaker than the conditions for regaining it. Consequently, actions which lower the trust state from its initial level and then rebuild it are less profitable for the responsible agent than maintaining high trust throughout. This cost increases with the depth and duration of the trust violation.

These properties are by no means exhaustive, and are intentionally quite broad, but they are intended to capture a certain range of behaviors which intuitively characterize trust, and which cannot be seen in memory-one strategies, or even in most long memory strategies. Properties **(T1)** and **(T2)** are also definitional properties of hysteresis, and distinguish trust-based strategies from mere retaliation. This reflects the experimentally observed distinctions between trust and reciprocity [23]. Property **(T3)** is of particular interest in establishing the deterrence value (and evolutionary value) of trust: If trust is known or assumed to be more expensive to rebuild than it is profitable to break, then there is a strong incentive to maintain a cooperative relationship. That this asymmetry is a fundamental component of trust is captured in the well-known phrase, "trust

takes years to build, and seconds to break." Without this property, properties **(T1)** and **(T2)** outline a more general class of state-dependent strategies with long memory.

The remainder of this paper is structured as follows: In Section 2, we will give some basic preliminaries and establish our notational conventions for the IPD. In Section 3, we will define the memory model used by this family of strategies, and we will define the elementary component of these models—the hysteron. In Section 4, we will define the family of four hysteron (4H) strategies, and will establish the geometric tools used to analyze their behavior, namely the memory square, threshold lines, and switching regions. In Section 5, we will use these tools to prove some useful propositions for analyzing the dynamics of these strategies, culminating in the deterrence theorem and its corollary, which provide conditions for the unprofitability of breaking trust (and hence the deterrent value of trust). In Section 6, we will address the problem of exploiting the 4H strategy, and provide results on the maximally exploitative cycles of moves which can be maintained indefinitely without breaking trust. As will be shown, this problem has deep connections to the combinatorics of words. Finally, in Section 8, we will summarize our main contributions, explore their evolutionary implications, and discuss some interesting open questions related to this family of strategies.

2 Preliminaries

While we assume the reader has a general familiarity with the prisoner's dilemma and its iterated counterpart, we will give some basic definitions here in the interest of establishing notation. The IPD consists of a series of rounds played by two players, denoted X and Y. In each round, X and Y must decide to either cooperate or defect. Cooperation is denoted by the letter C or the number 1, and defection is denoted by the letter D or the number 0. For brevity, when using the letter representation, we will write the pair of decisions of X and Y as CC, CD, DC, or DD, where the first and second letters represent the respective decisions of X and Y. When using the equivalent numerical representation, we will write it as the ordered pair $(1,1), (1,0), (0,1)$, or $(0,0)$. We denote the decisions made by X and Y on round n as X_n and Y_n, respectively.

2.1 Scoring

Each round confers a score or payoff to each player based on its outcome. These payoff values are denoted T, P, R, and S, usually taken to stand for "temptation payoff," "punishment payoff," "reward payoff," and "sucker's payoff." The payoff function π assigns one of these four payoff values to each player based on the round outcome. The IPD (as distinguished from other iterated games) is defined both by this assignment of payoffs to round outcomes, and also by an ordering condition on these values. In particular, the IPD requires

$$T > R > P > S, \quad \text{and} \quad 2T > T + S$$

and

$$\pi(X_n, Y_n) = \begin{cases} (R,R) & (X_n, Y_n) = (1,1), \\ (S,T) & (X_n, Y_n) = (1,0), \\ (T,S) & (X_n, Y_n) = (0,1), \\ (P,P) & (X_n, Y_n) = (0,0). \end{cases}$$

There are also the canonical payoff values

$$T = 5, \quad R = 3, \quad P = 1, \quad S = 0.$$

2.2 Strategies

We use the standard informal notions of a game history and a strategy, without introducing set-theoretic notation. A game history is a sequence of round outcomes; a strategy is a rule that assigns a decision to a game history. A strategy is said to be memory-n if it depends on n rounds of game history, or memory-∞ if it depends on an infinite game history. All strategies with memory must be initialized with initial conditions, which can be viewed as a "simulated" game history sufficient for the strategy to be able to make a decision on the first round of the game. The strategy family we define in this paper is naturally memory-∞, but can be directly approximated by a memory-n strategy for any n. When discussing the properties of these strategies, we will always take the perspective of the player X. We will assume that X uses a strategy from the family defined in this paper, and assume that Y is an opponent using an arbitrary strategy, which we may specify when necessary. The strategies we define have a specific memory structure, which we now describe.

2.3 Memory

We write $C = 1$ and $D = 0$, so that a decision can be treated as a binary digit. Then one can take the decision sequence $Y_1, \ldots, Y_n$ and concatenate the digits to obtain the binary word

$$Y_1 Y_2 \ldots Y_{n-1} Y_n.$$

We map this word to a binary representation of a number as follows:

$$m_n^Y = \frac{1}{2^n} m_0^Y + \sum_{i=1}^{n} \frac{1}{2^{n-i+1}} Y_i = \frac{1}{2} m_{n-1}^Y + \frac{1}{2} Y_n, \tag{1}$$

where $m_0^Y \in [0,1]$ is an initial condition. Then $m_n^Y \in [0,1]$, and m_n^Y can be thought of as representing an exponentially discounted memory of the decisions of Y, where the most recent decision has an equal weight to all previous decisions. Note that in this representation, we write 1 using its equivalent dyadic representation $0.111\ldots$, so that m_n^Y always begins with a 0 before the radix in floating point representation. Then the pair $m_n := (m_n^X, m_n^Y)$ denotes a history of n rounds, starting from some initial condition $m_0 := (m_0^X, m_0^Y)$, which we will call *initial memory*. We call $M := [0,1] \times [0,1]$ the *memory square*. Any point in the memory square can be interpreted as an initial condition m_0, or as the memory of a game history of length n together with an initial condition. Unless otherwise specified, whenever floating point representations of numbers are used throughout this paper, they are always assumed to be binary representations.

Remark 2.1. *The map from infinite game histories to M is not exactly bijective. Any number in $[0,1]$ with a finite binary representation (i.e. a dyadic rational) has two equivalent representations, e.g. $0.1000\cdots = 0.0111\ldots$, so the map is not injective for these histories. However, for the strategies we will define, two different representations of a dyadic rational are totally equivalent as initial conditions, in the sense that they will always lead to the same outcomes for any subsequent game history. This is due to the fact that these strategies depend directly on the value of $m \in M$ as a real number. Therefore, even though $0.1000\ldots$ and $0.0111\ldots$ differ on every move when viewed as game histories, they are entirely equivalent when viewed as initial conditions for our strategies. For this reason, we will consider M as uniquely representing all finite game histories taken with all distinct initial conditions, where two initial conditions $m_0, m_0' \in M$ are considered to be distinct if, for some finitely long continuation of the game, they lead to different outcomes.*

2.4 Behavioral terminology

Throughout this paper, we use the following descriptive behavioral terms: By *exploitation*, we mean defection by Y while X cooperates; this does not necessarily break trust. By *betrayal*, we mean defection by Y that is sufficient to break X's trust. By *punishment*, we mean defection by X while Y cooperates, following a betrayal. By *forgiveness*, we mean the return of X to a trusting state, following a period of punishment. What exactly we mean by "trust" and "breaking trust" will be developed in the following section.

3 Hysteresis

Hysteresis is a phenomenon which occurs in many models of physical processes such as ferromagnetism and elastoplastic deformation of materials [24, 25, 26]. It can be viewed as modeling "memory" in a very specific sense. Namely, it models a system whose output depends on its internal state, which in turn depends on extreme events in its past, such as extrema in its sequence of inputs. It is convenient building block of models for many natural processes where an input can produce a change, but reversing that input does not necessarily produce the reverse change. For example, applying a sufficiently strong magnetic field to an iron bar will magnetize it, but removing this field does not demagnetize it. Analogously in our application, a sequence of betrayals may break trust, but an equal sequence of cooperations may not be sufficient to rebuild trust, even if the sequence of round outcomes moves the final memory value m_n arbitrarily close to the initial memory value m_0.

First we will define the hysteron. Then we will define a complementary pair of elementary hysteretic strategies, called the "one hysteron" (1H) strategies. These strategies are not the main focus of this paper, but are included as a conceptually useful stepping stone to the "four hysteron" (4H) strategies which are our focus. We can also prove a simpler version of our main deterrence theorem in the context of these 1H strategies, which shows how the asymmetry property **(T3)** can appear in different trust-based strategies.

3.1 The Hysteron

The hysteron, also called the non-ideal relay, is an elementary building block of hysteresis operators. It is defined by two thresholds, $0 \leq \alpha \leq \beta$, and depends on an internal state $s \in \{0, 1\}$. Given an input $u \in \mathbb{R}$, its output $R_{\alpha,\beta}[s](u)$ is given by

$$R_{\alpha,\beta}[s](u) = \begin{cases} 1 & \text{if } u > \beta, \\ 0 & \text{if } u < \alpha, \\ s & \text{if } \alpha \leq u \leq \beta. \end{cases}$$

In the discrete time case, where s_n and x_n respectively denote the state and input at time n, $s_{n+1} = R_{\alpha,\beta}[s_n](x_n)$. The hysteron can be thought of as a simple model of a cooler controlled by a thermostat in a warm environment. If the temperature drops below α, the hysteron turns off the cooler, causing the temperature to rise. When the temperature exceeds β, the hysteron turns on the cooler, causing the temperature to fall. The space between α and β prevents the controller from repeatedly turning the cooler on and off as the temperature oscillates near the desired value, and this hysteresis is essential to the design of the system.

In our application, the state s_n is interpreted as whether the agent is *trusting* (when $s_n = 1$), or *distrusting* (when $s_n = 0$) on round n. The input is a memory value derived from the opponents

actions. Specifically, it is a convex combination of m^X and m^Y, the exponentially discounted memories of the agent's own actions and its opponent's actions. The thresholds α and β have a transparent behavioral interpretation in terms of binary expansions, which we develop in the context of the four hysteron strategy.

4 The four hysteron strategy

Any memory-one strategy can be described by a 4-tuple $p = (p_1, p_2, p_3, p_4)$, which gives the probabilities of cooperating if the previous round outcome was CC, CD, DC, and DD, respectively. For deterministic strategies, $p_i \in \{0, 1\}$, and for probabilistic strategies, $p_i \in [0, 1]$. However, a memory-one strategy cannot represent trust as a persistent state. If presented with an opponent's defection, it has no way of distinguishing whether this is a single defection following a long run of mutual cooperation, or the latest in an unbroken sequence of hundreds of consecutive defections. Without this larger contextual awareness, only basic reciprocity is possible, and sustained cooperation cannot be enforced or incentivized using an asymmetric cost. To give the strategy memory, path-dependence, and asymmetry, satisfying **(T1)**—**(T3)**, we replace each p_i with a hysteron, with independent thresholds, state, and input, whose state changes only when a threshold value is crossed based on an accumulated memory value of the agent's own actions and those of its opponent. We put

$$p_i = R_{\alpha_i,\beta_i}[s_{i,n}](u_{i,n}),$$

where $s_{i,n} \in \{0, 1\}$ for $i = 1, 2, 3, 4$ and $n \geq 0$, and where the inputs $u_{i,n}$ are convex combinations of m_n^X and m_n^Y of the form

$$\begin{aligned}
u_1(m_n^X, m_n^Y) &= u_{1,n} = \theta_1 m_n^X + (1 - \theta_1) m_n^Y, \\
u_2(m_n^X, m_n^Y) &= u_{2,n} = \theta_2 m_n^X + (1 - \theta_2) m_n^Y, \\
u_3(m_n^X, m_n^Y) &= u_{3,n} = \theta_3 m_n^X + (1 - \theta_3) m_n^Y, \\
u_4(m_n^X, m_n^Y) &= u_{4,n} = \theta_4 m_n^X + (1 - \theta_4) m_n^Y,
\end{aligned}$$

where $\theta_i \in [0, 1]$ is a weight which controls how much the i-th hysteron attends to X's own history versus Y's history. When $\theta_i = 1$, the input is m^X, so the hysteron responds only to X's own past actions; when $\theta_i = 0$, it responds only to Y's action. For each i, we use the update rule

$$u_{i,n+1} = \frac{1}{2} u_{i,n} + \frac{1}{2}\big(\theta_i X_{n+1} + (1 - \theta_i) Y_{n+1}\big).$$

This can simply be understood as the composition of the memory update rule 1 with the above input functions. We will refer to a strategy defined by these four hysterons with independent thresholds, states, and weights as a "four hysteron" (4H) strategy. As before, we also assume that $m_0^X, m_0^Y \in [0, 1]$ are chosen as initial conditions for each agent. The inputs and switching behaviors of the i-th hysteron can be understood in a unified way using the memory square M. If we write $u_{i,n}$ as a function $u_i(m)$, where $m \in M$ is taken as the n-round memory (m_n^X, m_n^Y) concatenated with the initial memory value (m_0^X, m_0^Y), then the loci of points such that $u_i(m) = \alpha_i$ and $u_i(m) = \beta_i$ form parallel lines in M. We denote the *lower threshold lines* L_{α_i} and the *upper threshold lines* L_{β_i}, defined

$$\begin{aligned}
L_{\alpha_i}, &:= \{(m^X, m^Y) \in M : \theta_i m^X + (1 - \theta_i) m^Y = \alpha_i\}, \\
L_{\beta_i}, &:= \{(m^X, m^Y) \in M : \theta_i m^X + (1 - \theta_i) m^Y = \beta_i\}.
\end{aligned}$$

Since the weights θ_i define convex combinations, these lines must have negative slope, if $\theta_i \in (0,1)$, or be vertical if $\theta_i = 1$, or horizontal if $\theta_i = 0$. The threshold lines also partition M into convex polygonal regions, and the switching regions are the open interiors of these regions (with the threshold lines themselves excluded[1]). It is also useful to label the switching regions, since if a trajectory enters this region, we can be certain that the associated state s_i changed. We define the *lower switching regions*

$$R_{\alpha_i} := \{(m^X, m^Y) \in M : u_i(m^X, m^Y) < \alpha_i\},$$

and the *upper switching regions*

$$R_{\beta_i} := \{(m^X, m^Y) \in M : u_i(m^X, m^Y) > \beta_i\}.$$

Then the definition of the hysteron clearly implies that, if $m_n \in R_{\alpha_i}$, then $s_i = 0$, and if $m_n \in R_{\beta_i}$, then $s_i = 1$.

To further aid this geometrical understanding, it is also useful to note how the four possible round outcomes map M into itself via affine contractions. Since $(X_n, Y_n) \in \{0,1\}^2$, each round outcome corresponds to one of four contractive affine transformations of the unit square. For a given history $m \in M$, we have

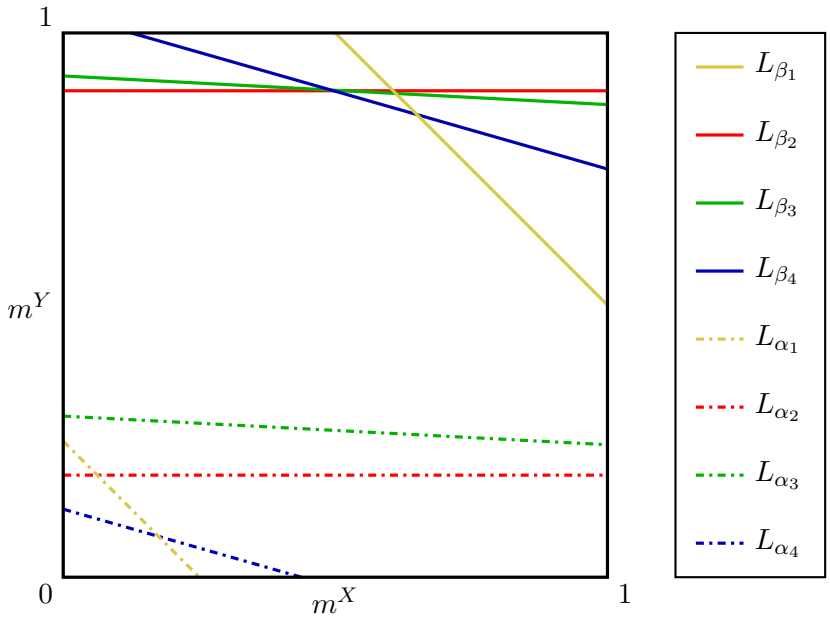
$$\begin{aligned}
F_{CC}(m_n) &:= \frac{1}{2}m_n + \left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}m_n^X + \frac{1}{2}, \frac{1}{2}m_n^Y + \frac{1}{2}\right), \\
F_{CD}(m_n) &:= \frac{1}{2}m_n + \left(\frac{1}{2}, 0\right) = \left(\frac{1}{2}m_n^X + \frac{1}{2}, \frac{1}{2}m_n^Y\right), \\
F_{DC}(m_n) &:= \frac{1}{2}m_n + \left(0, \frac{1}{2}\right) = \left(\frac{1}{2}m_n^X, \frac{1}{2}m_n^Y + \frac{1}{2}\right), \\
F_{DD}(m_n) &:= \frac{1}{2}m_n = \left(\frac{1}{2}m_n^X, \frac{1}{2}m_n^Y\right).
\end{aligned}$$

An immediate consequence of these four maps is that m_n is in the upper right quadrant $[1/2, 1] \times [1/2, 1]$ if and only if the most recent round outcome was CC, and likewise for the other three quadrants.

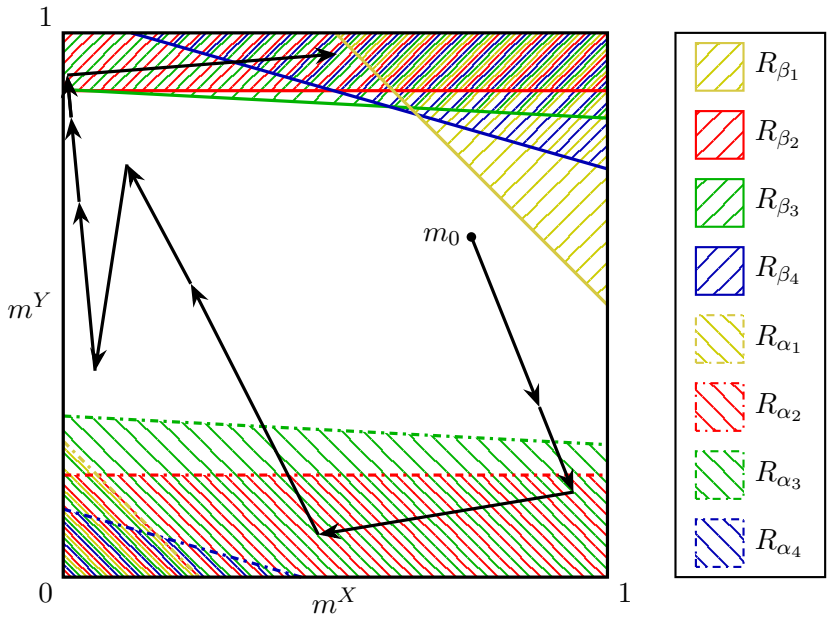
In summary, we have the following tools: The memory square M, which can be viewed as describing all game histories as well as all initial memory conditions. The thresholds α_i, β_i of each hysteron, and the weights θ_i of its associated input function, partition M into switching regions $R_{\alpha_i}, R_{\beta_i}$, bounded by threshold lines $L_{\alpha_i}, L_{\beta_i}$. Each round outcome maps the current memory value m_n to a new memory value m_{n+1} via one of the above four maps, which induces a trajectory through M starting from the initial memory value m_0. The states of the four hysterons change only if m_n enters one of the switching regions R_{α_i} or R_{β_i}, or, equivalently, if they cross the associated threshold lines. By analyzing these trajectories and obtaining conditions on the geometry and configuration of the switching regions, we can prove results about the behavior of the 4H strategy under a wide variety of different scenarios and across long time scales.

First of all, some conditions are necessary for X to even be capable of forgiveness. By "forgiveness" we generally mean when m_n enters R_{β_3} from a state where $s_{3,n} = s_{4,n} = 0$. If this is not possible, then X is said to be unforgiving, and this means that if X defects while $s_{3,n} = s_{4,n} = 0$, then X will defect on every subsequent round regardless of Y's decision.

[1]This is naturally because the hysteron, as defined here, only switches when $u_i < \alpha_i$ or $u_i > \beta_i$. Elsewhere in the literature, hysterons may be defined to switch on the boundary. In many applications this difference is immaterial, because the difference only exists on a space of inputs of zero measure. This is true here as well, generically; but some non-generic cases may be common or useful, e.g. starting from $m_0 = (1,1)$ and choosing $\alpha_i = 2^{-k}$.



(a) The memory square with threshold lines. Note that upper and lower threshold lines (paired here by color) are parallel, and all threshold lines must have negative slope (or be vertical or horizontal). Corresponding upper and lower threshold lines are drawn in the same color, but with different line styles.



(b) Here the switching regions are shaded, and we also show a trajectory of moves starting from the labeled m_0, assuming $s_{1,i} = 1$ for all i. In this case, Y's decisions are $DDDCCDCCCC$. X initially cooperates until the trajectory enters R_{α_2} after round 2, then X defects until the trajectory enters R_{β_3} on round 9.

Figure 1: The thresholds and weights, together with the initial memory and the initial states, are sufficient to determine the 4H strategy's response to any opponent strategy.

Proposition 4.1. *Let $s_{3,0} = s_{4,0} = 0$, $m_0 \in [0, \frac{1}{2}] \times [0, 1]$, and assume $m_0 \notin R_{\beta_3} \cup R_{\beta_4}$. Then X is unforgiving if*

$$\theta_3 \leq 2(\beta_3 - 1 + \theta_3)$$
$$\theta_4 \leq 2(\beta_4 - 1 + \theta_4)$$

Proof. Note that $m_0 \in [0, \frac{1}{2}] \times [0, 1]$ is equivalent to assuming that X defected in the most recent round. This is because the union of the ranges of F_{DC} and F_{DD} is exactly the left half-square $[0, \frac{1}{2}] \times [0, 1]$. Since we assume $m_0 \notin R_{\beta_3}$ and $m_0 \notin R_{\beta_4}$, and because the threshold lines bounding R_{β_3} and R_{β_4} must have negative slope, only a round outcome of DC can possibly map m_0 into R_{β_3} or R_{β_4}. Moreover, since $s_{4,0} = 0$ and $m_0 \notin R_{\beta_4}$, a DD round can only be followed by a DC round or a DD round, depending on the decision of Y. Therefore, if $R_{\beta_3} \cap [0, \frac{1}{2}] \times [\frac{1}{2}, 1] = \emptyset$ and $R_{\beta_4} \cap [0, \frac{1}{2}] \times [\frac{1}{2}, 1] = \emptyset$, it is impossible for either a DC or a DD round to map m_0 into R_{β_3} or R_{β_4}, and so X will always defect.

Note that if L_{β_3} or L_{β_4} intersect the left edge of M, then clearly $(R_{\beta_3} \cup R_{\beta_4}) \cap [0, \frac{1}{2}] \times [\frac{1}{2}, 1] \neq \emptyset$. So we can assume that L_{β_3} or L_{β_4} intersect the top edge of M (otherwise they do not intersect M at all, in which case X is clearly unforgiving). Since L_{β_3} has negative slope, it is therefore sufficient to see where it intersects the top edge of the memory square. Setting $m^Y = 1$, we obtain

$$\theta_3 m^X + (1 - \theta_3) = \beta_3,$$

which yields

$$m^X = \frac{\beta_3 - (1 - \theta_3)}{\theta_3}.$$

If $m^X < \frac{1}{2}$, then R_{β_3} intersects the upper left quadrant $[0, \frac{1}{2}] \times [\frac{1}{2}, 1]$. So we obtain

$$\frac{\beta_3 - 1 + \theta_3}{\theta_3} \geq \frac{1}{2},$$

assuming $\theta_3 > 0$, this rearranges to

$$\theta_3 \leq 2(\beta_3 - 1 + \theta_3).$$

In the event that $\theta_3 = 0$, then the lower boundary of R_{β_3} is given by $m^Y = \beta_3$, so $R_{\beta_3} \cap M = \emptyset$ iff $\beta_3 \geq 1$. This is equivalent to the above inequality if $\theta_3 = 0$, so this condition suffices for any $\theta_3 \in [0, 1]$. The argument for R_{β_4} proceeds identically. □

In a complementary way, we can establish conditions for the trust of X to be breakable at all. More specifically, we say X is always trusting if X will always cooperate given initial states $s_{1,0} = s_{2,0} = 1$ and initial memory $m_0 \notin R_{\alpha_1} \cup R_{\alpha_2}$. This means that, assuming X started in a cooperative state, then X will always cooperate regardless of the actions of Y.

Proposition 4.2. *Let $s_{1,0} = s_{2,0} = 1$, $m_0 \in [\frac{1}{2}, 1] \times [0, 1]$, and assume $m_0 \notin R_{\alpha_2} \cup R_{\alpha_2}$. Then X is always trusting if*

$$2\alpha_1 \leq \theta_1$$
$$2\alpha_2 \leq \theta_2$$

Proof. Assuming $m_0 \in [\frac{1}{2}, 1] \times [0, 1]$ is equivalent to assuming that X cooperated in the most recent round. The assumptions $s_{1,0} = s_{2,0} = 1$ and $m_0 \notin R_{\alpha_1} \cup R_{\alpha_2}$ likewise guarantee that the next

round outcome must be either CC or CD, depending on the action of Y. The remainder of the proof mirrors the proof of Proposition 4.1, but we will include it here because it is short, and demonstrates some of the symmetries of the memory square.

If L_{α_1} or L_{α_2} intersects the right edge of M, then clearly $(R_{\alpha_1} \cup R_{\alpha_2}) \cap [\frac{1}{2}, 1] \times [0, 1] \neq \emptyset$. So we assume that L_{α_1} and L_{α_2} intersect the bottom edge of M (or do not intersect M at all). Setting $m^Y = 0$, we obtain

$$\theta_1 m^X = \alpha_1.$$

If $m^X > \frac{1}{2}$, then L_{α_1} intersects the bottom right quadrant. So we obtain

$$2\alpha_1 \leq \theta_1.$$

We use the same logic to obtain the condition for α_2 and θ_2. □

The 4H strategies with the most interesting dynamics are those which are capable of forgiveness, and whose trust can be broken. Propositions 4.1 and 4.2 imply that such strategies must satisfy

$$\begin{aligned}
&\theta_1 < 2\alpha_1, \\
&\theta_2 < 2\alpha_2, \\
&2(\beta_3 - 1 + \theta_3) < \theta_3, \\
&2(\beta_4 - 1 + \theta_4) < \theta_4.
\end{aligned}$$

Next, we will give four useful results for determining the minimum number of successive rounds needed to enter any switching region from any starting point outside of that switching region. Note that due to the geometry of the regions and the affine maps, only F_{DC} and F_{CC} can send a point into the upper switching regions, and only F_{CD} and F_{DD} can send a point into the lower switching regions, which is reflected in the way the following four propositions are formulated.

Proposition 4.3. *Let $m_0 = (m_0^X, m_0^Y)$, and assume that $m_0 \notin R_{\alpha_i}$ for $i = 1, 2, 3, 4$. Then the minimum number of consecutive DD rounds needed for m_n to enter R_{α_i}, denoted $n = N_{DD}^{\alpha_i}$, is given by*

$$N_{DD}^{\alpha_i} = \left\lfloor \log_2 \left(\frac{\theta_i m_0^X + (1 - \theta_i) m_0^Y}{\alpha_i} \right) \right\rfloor + 1.$$

Proof. A round outcome of DD corresponds to the contractive affine map $(m^X, m^Y) \mapsto (\frac{1}{2}m^X, \frac{1}{2}m^Y)$. This moves (m_0^X, m_0^Y) along the line with slope $\frac{m_0^Y}{m_0^X}$, or along the vertical line connecting $(0, 0)$ and $(0, m_0^Y)$, if $m_0^X = 0$. We will first assume that $m_0^X \neq 0$. The minimum number of consecutive DD rounds to enter R_{α_i} depends on where this line intersects the threshold line $\theta_i m^X + (1-\theta_i) m^Y = \alpha_2$. Denoting the intersection point $(\hat{m}^X, \hat{m}^Y)$, it can be computed by solving the linear system

$$\begin{aligned}
\hat{m}^Y &= \frac{m_0^Y}{m_0^X} \hat{m}^X, \\
\theta_i \hat{m}^X + (1 - \theta_i) \hat{m}^Y &= \alpha_i,
\end{aligned}$$

which yields

$$\begin{aligned}
\hat{m}^X &= \frac{\alpha_i m_0^X}{\theta_i m_0^X + (1 - \theta_i) m_0^Y}, \\
\hat{m}^Y &= \frac{\alpha_i m_0^Y}{\theta_i m_0^X + (1 - \theta_i) m_0^Y}.
\end{aligned}$$

Since R_{α_i} is open, we need to move past the threshold line. This means that if $\log_2\left(\frac{m_0^Y}{\hat{m}^Y}\right) = k \in \mathbb{N}$, then after k compositions of the F_{DD} map, we will have $m_k = (\hat{m}^X, \hat{m}^Y)$, which is on the threshold line boundary of R_{α_i} but not in its interior. Therefore, the number of compositions of the map F_{DD} needed to move (m^X, m^Y) below $\hat{m}^Y$, or to the left of m_0^X, is given by

$$N_{DD}^{\alpha_i} = \left\lfloor \log_2\left(\frac{m_0^Y}{\hat{m}^Y}\right)\right\rfloor + 1 = \left\lfloor \log_2\left(\frac{m_0^X}{\hat{m}^X}\right)\right\rfloor + 1 = \left\lfloor \log_2\left(\frac{\theta_i m_0^X + (1-\theta_i)m_0^Y}{\alpha_i}\right)\right\rfloor + 1.$$

On the other hand, if $m_0^X = 0$, then the threshold line intersects the left edge of M at $(0, \frac{\alpha_i}{1-\theta_i})$, and so the minimum number of DD rounds is given by the same formula as above. □

Proposition 4.4. *Let $m_0 = (m_0^X, m_0^Y)$, and assume that $m_0 \notin R_{\alpha_i}$ for $i = 1, 2, 3, 4$. Then the minimum number of consecutive CD rounds needed for m_n to enter R_{α_i}, denoted $n = N_{CD}^{\alpha_i}$, is given by*

$$N_{CD}^{\alpha_i} = \left\lfloor \log_2\left(\frac{\theta_i(m_0^X - 1) + (1-\theta_i)m_0^Y}{\alpha_i - \theta_i}\right)\right\rfloor + 1.$$

Proof. A round outcome of CD corresponds to the contractive affine map $(m^X, m^Y) \mapsto (\frac{1}{2}m^X + \frac{1}{2}, \frac{1}{2}m^Y)$. If $m_0^X \neq 1$, this moves (m_0^X, m_0^Y) along the line with slope $\frac{m_0^Y}{m_0^X - 1}$. As in the previous proof, we will first assume that $m_0^X, m_0^Y \in (0, 0)$, and the same formula can be seen to apply in the cases where $m_0 = (1, 0)$ or $m_0 = (0, 1)$. The minimum number of consecutive CD rounds to enter R_{α_i} depends on where this line intersects the threshold line $\theta_i m^X + (1-\theta_i)m^Y = \alpha_i$. Again denoting the intersection point $(\hat{m}^X, \hat{m}^Y)$, we solve the linear system

$$\begin{aligned} \hat{m}^Y &= \frac{m_0^Y}{m_0^X - 1}(\hat{m}^X - 1) \\ \theta_i \hat{m}^X + (1-\theta_i)\hat{m}^Y &= \alpha_i, \end{aligned}$$

obtaining

$$\begin{aligned} \hat{m}^X &= \frac{(m_0^X - 1)(\alpha_i - \theta_i)}{\theta_i(m_0^X - 1) + (1-\theta_i)m_0^Y} + 1, \\ \hat{m}^Y &= \frac{m_0^Y(\alpha_i - \theta_i)}{\theta_i(m_0^X - 1) + (1-\theta_i)m_0^Y}. \end{aligned}$$

For the purposes of this problem, it is useful to note that the map $m^X \mapsto \frac{1}{2}m^X + \frac{1}{2}$ is equivalent to the map sending $1 - m^X \mapsto \frac{1}{2}(1 - m^X)$, in the sense that each composition of F_{CD} halves the distance between m^X and 1. Then the number of compositions of the map F_{CD} needed to move (m^X, m^Y) below $\hat{m}^Y$ (and therefore into R_{α_i}) is given by

$$\begin{aligned} N_{CD}^{\alpha_i} &= \left\lfloor \log_2\left(\frac{m_0^Y}{\hat{m}^Y}\right)\right\rfloor + 1 = \left\lfloor \log_2\left(\frac{1 - m_0^X}{1 - \hat{m}^X}\right)\right\rfloor + 1 \\ N_{CD}^{\alpha_i} &= \left\lfloor \log_2\left(\frac{\theta_i(m_0^X - 1) + (1-\theta_i)m_0^Y}{\alpha_i - \theta_i}\right)\right\rfloor + 1 \end{aligned}$$

□

Since CC and DC round outcomes can never cause trajectories to enter lower switching regions, the following two propositions will instead concern the upper switching regions.

Proposition 4.5. *Let $m_0 = (m_0^X, m_0^Y)$, and assume that $m_0 \notin R_{\beta_i}$ for $i = 1, 2, 3, 4$. Then the minimum number of consecutive CC rounds needed for m_n to enter R_{β_i}, denoted $n = N_{CC}^{\alpha_i}$, is given by*

$$N_{CC}^{\beta_i} = \left\lfloor \log_2 \left(\frac{\theta_i(1 - m_0^X) + (1 - \theta_i)(1 - m_0^Y)}{1 - \beta_i} \right) \right\rfloor + 1$$

Proof. A round outcome of CC corresponds to the map $(m^X, m^Y) \mapsto (\frac{1}{2}m^X + \frac{1}{2}, \frac{1}{2}m^Y + \frac{1}{2})$, moving (m_0^X, m_0^Y) along the line with slope $\frac{m_0^Y}{m_0^X - 1}$. As in the previous proofs, we will first assume that $m_0^X, m_0^Y \in (0, 0)$, but the final result also applies in the cases where $m_0 = (1, 0)$ or $m_0 = (0, 1)$. Finally, we will also note the fact (also mentioned towards the end of the proof of Proposition 4.4) that the number of compositions of the affine map needed for $m_n^X > \hat{m}^X$ is equivalent to the number of compositions of $1 - m_0^X \mapsto \frac{1}{2}(1 - m_0^X)$ needed for $1 - m_n^X < 1 - \hat{m}_0^X$, as can be easily seen. This important symmetry can be seen in a comparison of the statements and proofs of Propositions 4.3 and 4.4, and can also be exploited here.

Again denoting the intersection point $(\hat{m}^X, \hat{m}^Y)$, we obtain the linear system

$$\begin{aligned} \hat{m}^Y &= \frac{m_0^Y - 1}{m_0^X - 1}(\hat{m}^X - 1) + 1, \\ \theta_i \hat{m}^X + (1 - \theta_i)\hat{m}^Y &= \beta_i. \end{aligned}$$

The symmetries of this system to the one shown in the proof of Proposition 4.3 can be easily seen by rewriting the above as

$$\begin{aligned} (1 - \hat{m}^Y) &= \frac{1 - m_0^Y}{1 - m_0^X}(1 - \hat{m}^X), \\ \theta_i(1 - \hat{m}^X) + (1 - \theta_i)(1 - \hat{m}^Y) &= 1 - \beta_i. \end{aligned}$$

It is useful to solve these in terms of $1 - \hat{m}^X$ and $1 - \hat{m}^Y$, obtaining

$$\begin{aligned} 1 - \hat{m}^X &= \frac{(1 - \beta_i)(1 - m_0^X)}{\theta_i(1 - m_0^X) + (1 - \theta_i)(1 - m_0^Y)}, \\ 1 - \hat{m}^Y &= \frac{(1 - \beta_i)(1 - m_0^Y)}{\theta_i(1 - m_0^X) + (1 - \theta_i)(1 - m_0^Y)}. \end{aligned}$$

Then the number of compositions of the map F_{CC} needed to move (m^X, m^Y) into R_{β_i} is given by

$$\begin{aligned} N_{CC}^{\beta_i} &= \left\lfloor \log_2 \left(\frac{1 - m_0^Y}{1 - \hat{m}^Y} \right) \right\rfloor + 1 = \left\lfloor \log_2 \left(\frac{1 - m_0^X}{1 - \hat{m}^X} \right) \right\rfloor + 1 \\ N_{CC}^{\beta_i} &= \left\lfloor \log_2 \left(\frac{\theta_i(1 - m_0^X) + (1 - \theta_i)(1 - m_0^Y)}{1 - \beta_i} \right) \right\rfloor + 1 \end{aligned}$$

□

Proposition 4.6. *Let $m_0 = (m_0^X, m_0^Y)$, and assume that $m_0 \notin R_{\beta_i}$ for $i = 1, 2, 3, 4$. Then the minimum number of consecutive DC rounds needed for m_n to enter R_{β_i}, denoted $n = N_{CC}^{\alpha_i}$, is given by*

$$N_{DC}^{\beta_i} = \left\lfloor \log_2 \left(\frac{\theta_i m_0^X + (1 - \theta_i)(m_0^Y - 1)}{\beta_i + \theta_i - 1} \right) \right\rfloor + 1$$

Proof. A round outcome of DC corresponds to the map $(m^X, m^Y) \mapsto (\frac{1}{2}m^X, \frac{1}{2}m^Y + \frac{1}{2})$, moving (m_0^X, m_0^Y) along the line with slope $\frac{m_0^Y}{m_0^X - 1}$. As before, we assume that $m_0^X, m_0^Y \in (0,0)$, but the final result also applies in the cases where $m_0 = (1,0)$ or $m_0 = (0,1)$. We again use the same symmetries. Putting $(\hat{m}^X, \hat{m}^Y)$ for the intersection of the trajectory line of the affine map with the upper threshold line, we obtain the linear system

$$\begin{aligned} \hat{m}^Y &= \frac{m_0^Y - 1}{m_0^X}\hat{m}^X + 1, \\ \theta_i \hat{m}^X + (1-\theta_i)\hat{m}^Y &= \beta_i. \end{aligned}$$

From this, we obtain

$$\begin{aligned} \hat{m}^X &= \frac{(\beta_i + \theta_i - 1)m_0^X}{\theta_i m_0^X + (1-\theta_i)(m_0^Y - 1)}, \\ \hat{m}^Y &= \frac{(\beta_i + \theta_i - 1)(m_0^Y - 1)}{\theta_i m_0^X + (1-\theta_i)(m_0^Y - 1)} + 1. \end{aligned}$$

Then the number of compositions of the map F_{DC} needed to move (m^X, m^Y) into R_{β_i} is given by

$$\begin{aligned} N_{DC}^{\beta_i} &= \left\lfloor \log_2 \left(\frac{1 - m_0^Y}{1 - \hat{m}^Y} \right) \right\rfloor + 1 = \left\lfloor \log_2 \left(\frac{m_0^X}{\hat{m}^X} \right) \right\rfloor + 1 \\ N_{DC}^{\beta_i} &= \left\lfloor \log_2 \left(\frac{\theta_i m_0^X + (1-\theta_i)(m_0^Y - 1)}{\beta_i + \theta_i - 1} \right) \right\rfloor + 1 \end{aligned}$$

□

These four propositions can be used to study many types of switching behavior starting from any game state. Of particular interest is breaking and rebuilding trust, which we will call a *trust cycle*, and define in the following section.

5 Trust cycles

In order to give a formal definition of a trust cycle in terms of trajectories on M, it is necessary to make a few assumptions on the relationship of the various switching regions. We call the following assumptions the "no shortcut" assumptions

(NS1) $R_{\alpha_2} \subseteq R_{\alpha_3}$,

(NS2) $R_{\alpha_2} \subseteq R_{\alpha_4}$,

(NS3) $R_{\beta_3} \subseteq R_{\beta_2}$,

(NS4) $R_{\beta_4} \subseteq R_{\beta_2}$.

Without these assumptions, it is possible for a trajectory m_n to enter R_{α_2} without having first entered R_{α_3} or R_{α_4}. Indeed, suppose that the trajectory m_n enters R_{α_2} on round n without having entered R_{α_3} or R_{α_4}, and suppose the initial states were $s_{1,0} = s_{2,0} = s_{3,0} = s_{4,0} = 1$. In this case, X will defect after the CD round which sent $m_n \in R_{\alpha_2}$. If Y cooperates on round $n+1$, then the outcome on round $n+1$ will be DC. If m_{n+1} did not enter R_{α_3}, then $s_{3,n+1} = 1$, so round $n+2$ will be CC, which could be repeated indefinitely to immediately rebuild trust. On

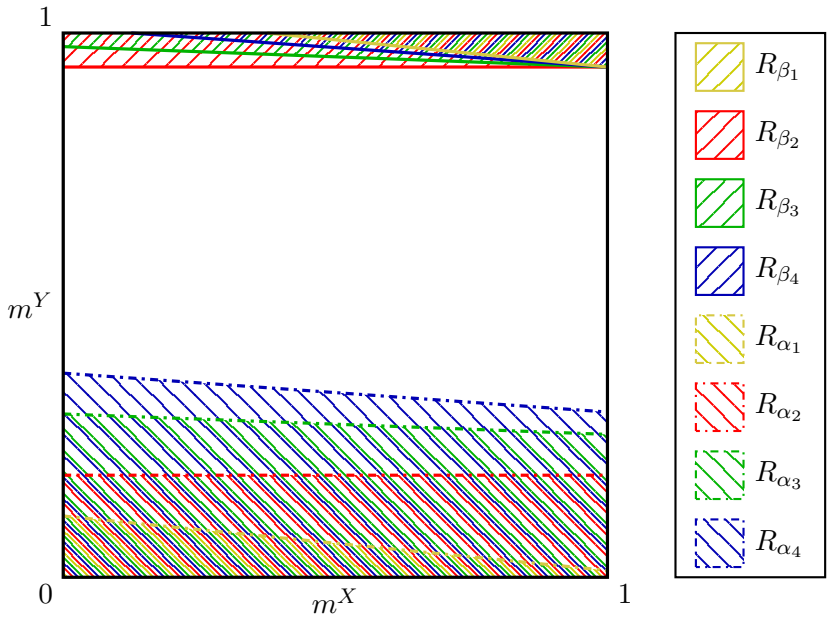


Figure 2: An example of thresholds and switching regions satisfying the no shortcut assumptions **(NS1)**–**(NS4)**.

the other hand, if Y defected on round $n+1$, then round $n+1$ will be DD. If m_{n+1} did not enter R_{α_4}, then Y can choose to cooperate on round $n+2$, again leading to CC, which could be repeated to rebuild trust. Viewing these scenarios as "shortcuts" bypassing the normal trust rebuilding process, we forbid them in order to force trajectories that break and rebuild trust to follow the $CD \to \cdots \to CD \to DC \to \cdots \to DC$ geometry shown in Figure 3. We make the same assumptions on the upper switching regions for the same reasons.

Remark 5.1. *There are, of course, other ways to prevent these undesirable transitions from occurring. For example, one might instead require that $s_{3,0} = s_{4,0} = 0$, which ensures that X will continue defecting after defecting once (e.g. on the round after m_n enters R_{α_2}). However, this approach implies some necessary conditions on the initial memory $m_0 = (1,1)$, namely that m_0 must lie outside $R_{\beta_3} \cap R_{\beta_4}$. Otherwise, if $m_0 \in R_{\beta_3} \cup R_{\beta_4}$, this contradicts $s_{3,0} = 0$ or $s_{4,0} = 0$. This makes it difficult to consider the full trust start where $m_0 = (1,1)$, which represents the best case scenario for an exploitative player Y. The no-shortcut assumptions, though slightly more complicated on their surface, let us sidestep this inconsistency.*

We will also assume that the initial state $s_{2,0} = 1$. The values of the other initial states do not affect the following analysis. Under these assumptions, we formally define a trust cycle as a trajectory with $m_0 \in M \setminus (\bigcup_{i=1}^4 R_{\alpha_i})$, which enters R_{α_2} on some round n_1, and which enters R_{β_3} on some round $n_2 > n_1$. We consider the first round where $m_{n_2} \in R_{\beta_3}$ to be the end of the trust cycle. Trust cycles are characterized by the way X responds to defections from Y, and can be decomposed into two legs. The first leg, which we call the *betrayal leg*, comprises rounds $1, \ldots, n_1$, ending when $m_{n_1} \in R_{\alpha_2}$ for the first time. During the betrayal leg, X cooperates in response to Y's defections. The second leg, which we call the *punishment leg*, comprises rounds $n_1+1, \ldots, n_2$. Its end coincides with the end of the cycle, when trust is regained. During the punishment leg, X defects in response to Y's cooperations (in fact, under the above assumptions, X defects during the punishment leg no matter what Y does).

For the purpose of studying the asymmetric cost of trust and its deterrence value, we will assume that Y defects continuously during the betrayal leg, and cooperates continuously during the punishment leg. As we will see in the second part of this paper, Y can actually be much more exploitative by interleaving cooperations and defections in certain optimal periodic sequences.

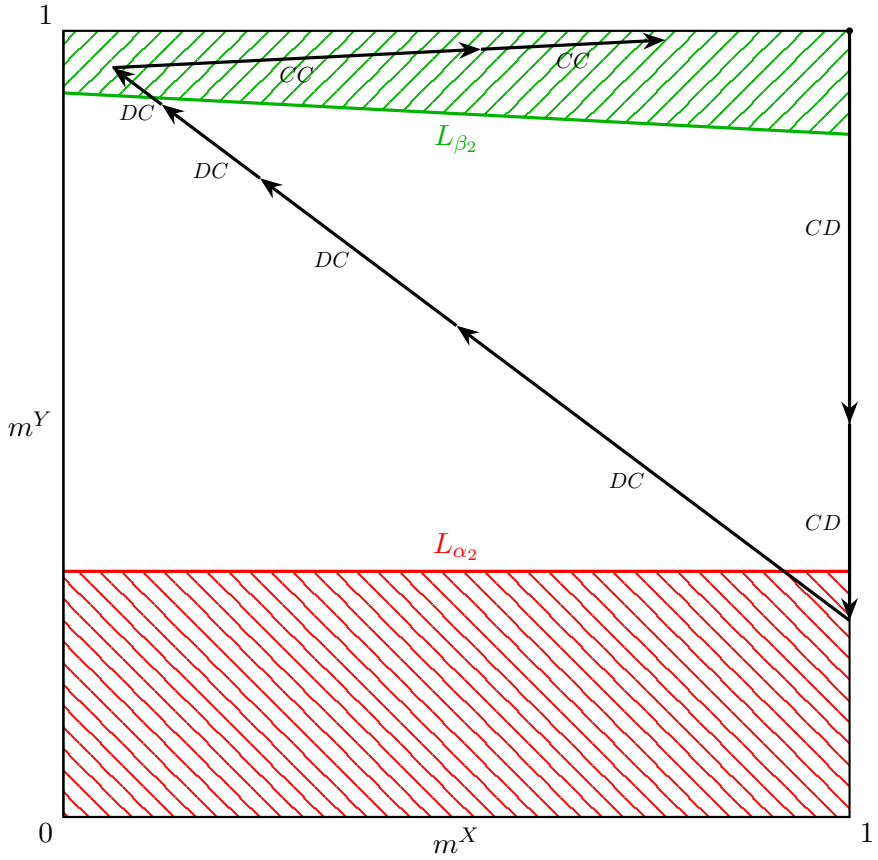


Figure 3: An example trust cycle starting from $m_0 = (1, 1)$. Here, $\alpha_2 = \frac{5}{16}$, $\theta_2 = 0$, $\beta_3 = \frac{7}{8}$, and $\theta_3 = \frac{1}{20}$. The red and green shaded regions represent R_{α_2} and R_{β_3}, and the other switching regions and threshold lines have been omitted for clarity. The betrayal leg consists of 2 rounds, and the punishment leg is 4 rounds. Note the characteristic triangular shape, which would asymptotically approach a closed loop if CC rounds continued.

In order to find a lower bound on the cost of trust, we will assume $m_0 = (1, 1)$. This is a "full trust" start where X sees a history of perfect cooperation from itself and from Y, which gives Y the largest possible number of defections (i.e. the longest possible betrayal leg) before the trajectory enters R_{α_2} and X begins retaliating. For the same reason, we will also assume that after the trajectory enters R_{α_2}, Y immediately begins cooperating on the following round. In a real game where Y does not know the trust thresholds of X, this is unlikely, because Y would not know that it has broken X's trust until X begins defecting in the following round. The resulting mutual defection round would push the trajectory towards the bottom left corner, further increasing the distance to the R_{β_3} switching region and lengthening the punishment leg. So, our analysis is based on the best possible case for the exploiter Y.

Propositions 4.4 and 4.6 give the necessary tools to compute the leg lengths, but it is also useful to first obtain some very simple corollaries based on the above assumptions.

Corollary 5.1. *If $m_0 = (1,1)$, then the minimum number of CD rounds needed for $m_n \in R_{\alpha_2}$ is given by*

$$N_{CD}^{\alpha_2} = \lfloor -\log_2(\tau_{\alpha_2}) \rfloor + 1,$$

and $m_n = \left(1, 2^{-N_{CD}^{\alpha_2}}\right)$

Proof. This follows immediately from Proposition 4.4. □

Corollary 5.2. *If $m_n = \left(1, 2^{-N_{CD}^{\alpha_2}}\right)$, the minimum number of DC rounds needed for $m_{n+k} \in R_{\beta_3}$, denoted $k = N_{DC}^{\beta_3}$, satisfies*

$$N_{DC}^{\beta_3} \geq \left\lfloor \log_2 \left(\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \right) \right\rfloor + 1,$$

and

$$N_{DC}^{\beta_3} \leq \left\lfloor \log_2 \left(\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 + \theta_2 - 2)}{2(\beta_3 + \theta_3 - 1)(1-\theta_2)} \right) \right\rfloor + 1,$$

Proof. Assume $m_0 = (1,1)$ and let n denote the first round where $m_n \in R_{\alpha_2}$. Since R_{α_2} is open, $m_n^Y \in [\frac{1}{2}\tau_{\alpha_2}, \tau_{\alpha_2})$. By Proposition 4.6, we have

$$\log_2 \left(\frac{1-\tau_{\alpha_2}}{1-\hat{m}^Y} \right) < \log_2 \left(\frac{1-m_n^Y}{1-\hat{m}^Y} \right) \leq \log_2 \left(\frac{1-\frac{\tau_{\alpha_2}}{2}}{1-\hat{m}^Y} \right),$$

so

$$\left\lfloor \log_2 \left(\frac{1-\tau_{\alpha_2}}{1-\hat{m}^Y} \right) \right\rfloor + 1 \leq N_{DC}^{\beta_3} \leq \left\lfloor \log_2 \left(\frac{1-\frac{\tau_{\alpha_2}}{2}}{1-\hat{m}^Y} \right) \right\rfloor + 1.$$

Using the definition of τ_{α_2} and of $\hat{m}^Y$ from Proposition 4.6, the result follows. □

These results allow us to formulate the main deterrence theorem, which establishes the asymmetry property of trust.

Theorem 5.1. *Suppose $m_0 = (1,1)$, and the parameters of the 4H strategy of X satisfy **(NS1)**–**(NS4)**. Let m_n denote a trust cycle trajectory where Y defects continuously until m_i enters R_{α_2} on round $i < n$, then cooperates continuously until m_n enters R_{β_3} on round n. Let $\Delta\pi(m_n) = P^X(m_n) - P^Y(m_n)$ denote the difference in cumulative payoff on round n between X and Y. Then*

$$\Delta\pi(m_n) = (T-S)(N_{DC}^{\beta_3} - N_{CD}^{\alpha_2}),$$

where $T - S$ is the payoff difference for X on a DC round, and where

$$N_{DC}^{\beta_3} - N_{CD}^{\alpha_2} = \left\lfloor \log_2 \left(\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \right) \right\rfloor - \left\lfloor \log_2 \left(\frac{1-\theta_2}{\alpha_2 - \theta_2} \right) \right\rfloor.$$

Proof. On a single CD round, the value of $\Delta\pi$ changes by $S - T$, and on a single DC round, $\Delta\pi$ changes by $T - S$. Therefore, we have

$$\Delta\pi = (T-S)(N_{DC}^{\beta_3} - N_{CD}^{\alpha_2}).$$

By Corollary 5.1, there are $N_{CD}^{\alpha_2} = \lfloor -\log_2(\tau_{\alpha_2}) + 1 \rfloor = \left\lfloor \log_2 \left(\frac{1-\theta_2}{\alpha_2 - \theta_2} \right) + 1 \right\rfloor$ of these rounds. Since we are seeking a lower bound $\Delta\pi$, we assume that $m_i^Y - \tau_{\alpha_2} = \varepsilon > 0$ where ε is sufficiently small

(i.e. that m_i entered R_{α_2} arbitrarily close to the threshold line). Then the number of DC rounds needed to rebuild trust is given by Corollary 5.2 as

$$N_{DC}^{\beta_3} = \left\lfloor \log_2 \left(\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \right) + 1 \right\rfloor,$$

and the result follows immediately. □

Corollary 5.3. *Under the same assumptions of Theorem 5.1, $\Delta\pi(m_n) > 0$ if the threshold parameters satisfy*

$$\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \geq 2\frac{1-\theta_2}{\alpha_2 - \theta_2}.$$

Moreover, if $\theta_2 = \theta_3 = 0$, a lower bound for α_2 in terms of β_3 is given by

$$\alpha_2 \geq \frac{1}{2} - \sqrt{\frac{1}{4} + 2(\beta_3 - 1)},$$

a lower bound for β_3 in terms of α_2 is given by

$$\beta_3 \geq 1 - \frac{\alpha_2(1-\alpha_2)}{2},$$

and $\left\lfloor \log_2 \left(\frac{\alpha_2(1-\alpha_2)}{1-\beta_2} \right) \right\rfloor$ is a sharp lower bound for $N_{DC}^{\beta_3} - N_{CD}^{\alpha_2}$, which differs from it by at most 1, i.e.

$$\left\lfloor \log_2 \left(\frac{\alpha_2(1-\alpha_2)}{1-\beta_2} \right) \right\rfloor \leq N_{DC}^{\beta_3} - N_{CD}^{\alpha_2} \leq \left\lfloor \log_2 \left(\frac{\alpha_2(1-\alpha_2)}{1-\beta_2} \right) \right\rfloor + 1$$

Proof. By Theorem 5.1, $\Delta\pi(m_n) > 0$ if

$$\left\lfloor \log_2 \left(\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \right) + 1 \right\rfloor - \left\lfloor \log_2 \left(\frac{1-\theta_2}{\alpha_2 - \theta_2} \right) + 1 \right\rfloor > 0,$$

which holds if

$$\left\lfloor \log_2 \left(\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \right) + 1 \right\rfloor > \left\lfloor \log_2 \left(\frac{1-\theta_2}{\alpha_2 - \theta_2} \right) + 1 \right\rfloor.$$

A sufficient condition for the above is that

$$\log_2 \left(\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \right) - \log_2 \left(\frac{1-\theta_2}{\alpha_2 - \theta_2} \right) \geq 1.$$

Combining logs and exponentiating, we obtain

$$\frac{\theta_3(1-\theta_2) + (1-\theta_3)(\alpha_2 - 1)}{(\beta_3 + \theta_3 - 1)(1-\theta_2)} \geq 2\frac{1-\theta_2}{\alpha_2 - \theta_2}.$$

From here, setting $\theta_2 = \theta_3 = 0$, we obtain

$$\frac{\alpha_2 - 1}{\beta_3 - 1} \geq \frac{2}{\alpha_2}.$$

By manipulating this inequality, we obtain the given conditions for α_2 in terms of β_3, or β_3 in terms of α_2. Finally, by using this inequality and considering the floor functions in the statement of Theorem 5.1, we can see that

$$\log_2 \left(\frac{\alpha_2(1-\alpha_2)}{1-\beta_2} \right)$$

can differ from $N_{DC}^{\beta_3} - N_{CD}^{\alpha_2}$ by at most 1, and serves as a sharp lower bound on the difference of the lengths of the two legs (which is proportional to the payoff difference $\Delta\pi$, scaled by a factor of the net temptation payoff value $T - S$). □

Remark 5.2. *The deterrence theorem and its corollary have a direct evolutionary interpretation: Strategies that tolerate betrayal without punishment are vulnerable to exploitation, while strategies that punish too harshly risk permanent mutual defection. Trust-based strategies, particularly 4H strategies satisfying Corollary 5.3 navigate between these extremes by making betrayal unprofitable while preserving the possibility of reconciliation. We do not test the evolutionary consequences of this result here, but the payoff asymmetry it establishes would be well-suited to a replicator or invasion analysis.*

6 The exploiter's optimization problem

Suppose X is trusting, and Y wishes to exploit X. By exploitation, we mean that Y wants to defect while X cooperates. Assuming that breaking trust incurs a net negative payoff for Y, Y wants to find a sequence of moves which contains as many defections as possible without breaking trust. We will view such a sequence of n moves as a binary word $y_n = Y_1 Y_2 \dots Y_n$, and we will write the truncation of y_n to its first k moves as $y_k = Y_1 Y_2 \dots Y_k$ for $k < n$. These words have a natural lexicographic order, which coincides with their order as binary representations of integers, and if we write $y_n > w_n$, this comparison is to be understood in terms of lexicographic order. When using a subscript to denote the length of a word is inconvenient, we may simply write y and denote its length by $|y|$, and we will use $|y|_1$ and $|y|_0$ to denote the number of 1s and 0s in y, respectively. We will write concatenation of words using multiplicative notation, so y_n^q denotes q repetitions of y_n. Similarly, if y_n can be written $y_n = uv$ where u and v are subwords of y, this is called a *factorization* of y, and u and v are factors of y. A *rotation*[2] of y_n is written $r_k(y_n)$, and is a cyclic permutation of its elements mapping $Y_i \mapsto Y_{i+k}$ with indices taken modulo n, i.e.

$$r_k(y_n) = Y_{k+1} Y_{k+2} \dots Y_n Y_1 Y_2 \dots Y_k.$$

Finally, we will write $\tilde{y}_n$ for the reversal of y_n, i.e.

$$\tilde{y}_n = Y_n Y_{n-1} \dots Y_1.$$

Assuming $m_0^Y = 1$, we will write the m^Y component of the trajectory induced by y_n, starting from m_0^Y, as $m^Y(y_n)$. We can now define two essential quantities for this optimization problem. The first is the *density* ρ, defined

$$\rho(y_n) := \frac{1}{n} \sum_{i=1}^{n} Y_i.$$

This represents the proportion of the total move sequence which is made up of cooperations. Second, given a periodic word $y = y_n y_n y_n \dots$, we define the *cyclic minimum trust value* $\tau_{\min}(y_n)$ as the minimum value of m^Y on the limit cycle of y_n. Equivalently,

$$\tau_{\min}(y_n) := \lim_{r \to \infty} \min_{k \leq rn} m^Y(y_n^r).$$

A closed form expression for $\tau_{\min}$ is given in the following lemma:

[2] Also commonly called a conjugation in the literature of combinatorics on words. We will use the term "rotation" for consistency's sake.

Lemma 6.1. *Given a move sequence y_n, its cyclic minimum trust value $\tau_{\min}(y_n)$ is given by*

$$\tau_{\min}(y_n) = \frac{\min_{1\leq k\leq n} \sum_{i=1}^{n} 2^{i-1} Y_{k+i}}{2^n - 1}$$

Proof. A limit cycle of $m^Y(y_n^r)$ as $r \to \infty$ consists of n distinct points. To see this, consider the memory map $m^Y \mapsto \frac{1}{2}m^Y + \frac{1}{2}Y_1$. Its n-fold composition is given by

$$m^Y \mapsto \frac{1}{2^n} m^Y + \sum_{i=1}^{n} \frac{1}{2^{n-i+1}} Y_i.$$

Since this is a contractive affine map on M, it must have a fixed point by the Banach fixed point theorem. A fixed point of this map must satisfy

$$\begin{aligned}
m^Y &= \frac{1}{2^n} m^Y + \sum_{i=1}^{n} \frac{1}{2^{n-i+1}} Y_i, \\
2^n m^Y &= m^Y + \sum_{i=1}^{n} 2^{i-1} Y_i, \\
(2^n - 1) m^Y &= \sum_{i=1}^{n} 2^{i-1} Y_i, \\
m^Y &= \frac{\sum_{i=1}^{n} 2^{i-1} Y_i}{2^n - 1}.
\end{aligned}$$

Since $Y_{i+n} = Y_i$, this equation has n distinct solutions, corresponding to the n rotations of y_n. Taking the minimum value of these fixed points, the result follows. □

Note that for a periodic sequence made from a period n cycle given by y_n, the quantities $\rho(y_n)$ and $\tau_{\min}(y_n)$ are invariant under rotation.

The exploiter's optimization problem is to find a sequence of n moves y_n, which minimizes $\rho(y_n)$ while maximizing $\tau(y_n)$. We will call such a sequence a *maximally exploitative sequence*. If y_n is taken as the fundamental period of a periodic sequence, we similarly call y_n a *maximally exploitative cycle* if it minimizes $\rho(y_n)$ while maximizing $\tau_{\min}(y_n)$. The maximally exploitative cycle is the periodic cycle of cooperations and defections which can be sustained indefinitely without breaking trust, and which has the lowest density. Since maximally exploitative cycles determine long-term behavior, and differ from maximally exploitative sequences only for finitely many (usually very few) initial moves away from the limit cycle, maximally exploitative cycles will be the focus of this section. As it turns out, the problem of finding maximally exploitative cycles has deep connections to the combinatorics of words. We will give a very brief and self-contained introduction to the necessary concepts and definitions throughout this section as needed.

6.1 Maximally exploitative cycles and Christoffel words

First, we will give the classical definition of the lower Christoffel words from the literature [27, 28]. Take a point $(a, b) \in \mathbb{Z}\times\mathbb{Z}$ with $\gcd(a, b) = 1$, and consider the line segment with slope $\frac{b}{a}$ connecting $(0, 0)$ and (a, b). The *lower Christoffel path* of slope $\frac{b}{a}$ is the unique path on the integer lattice $\mathbb{Z} \times \mathbb{Z}$ satisfying the following two conditions:

1. The path lies below the line segment connecting $(0, 0)$ and (a, b).

2. The region in the plane enclosed by the path and the line segment contains no other points of $\mathbb{Z} \times \mathbb{Z}$ besides those on the path.

In other words, it is the path composed of horizontal and vertical line segments of unit length connecting $(0, 0)$ and (a, b) which lies below and is closest to the line segment connecting these two points. Since each segment of the path connecting two integer coordinates has unit length and is either horizontal or vertical, we can encode the horizontal steps as 0 and the vertical steps as 1, and obtain a binary word of length $a + b$ by concatenating the segments along the lower Christoffel path from $(0, 0)$ to (a, b). This is called the *lower Christoffel word* of slope $\frac{b}{a}$. The upper Christoffel path and upper Christoffel word are defined analogously, except the path is taken above the line segment. Henceforth, when we say "Christoffel word" without additional qualification, we will always mean the lower Christoffel word.

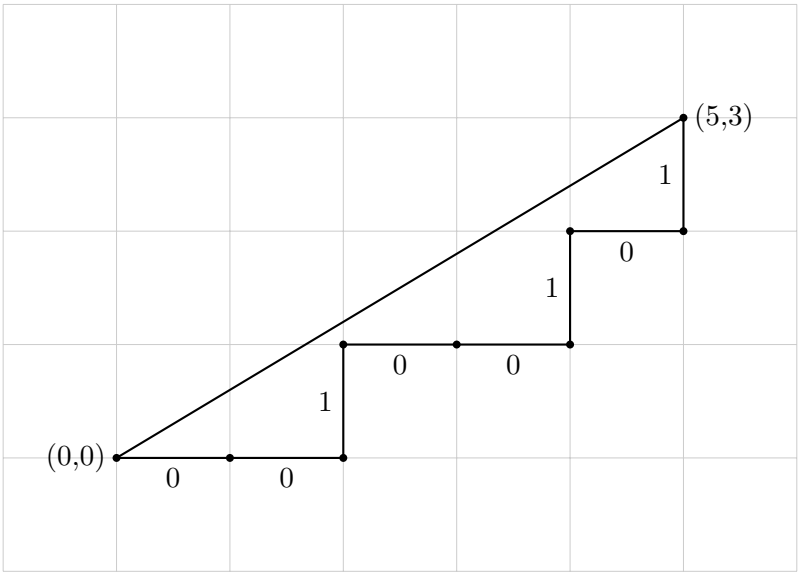


Figure 4: The lower Christoffel path of slope $\frac{3}{5}$, which corresponds to the lower Christoffel word $c(5, 3) = 00100101$, which has density $\frac{3}{8}$. This figure follows the standard presentation in [28].

We will denote the Christoffel word with slope $\frac{b}{a}$ as $c(a, b)$. The slope of a Christoffel word and its density as a sequence of moves are closely related. Clearly, $\rho(c(a, b)) = \frac{b}{a+b}$, and a Christoffel word with density $\rho = \frac{p}{q}$ will have slope $\frac{p}{q-p}$, and can be written $c(q - p, p)$.

Lemma 6.2. *Given $n = a + b$ with $\gcd(a, b) = 1$, if $y_n = c(a, b)$, then*

$$\tau_{\min}(y_n) = \frac{\sum_{i=1}^{n} 2^{n-i} Y_i}{2^n - 1}.$$

Proof. First note that numerator $\sum_{i=1}^{n} 2^{n-i} Y_i$ is y_n, interpreted as a binary representation of an integer, and $\tau_{\min}$ is defined as the rotation of $\tilde{y}_n$ which is minimal as a binary representation of an integer. This is equivalent to the rotation of $\tilde{y}_n$ which has the smallest lexicographic value. So, finding $\tau_{\min}(y_n)$ is equivalent to finding the rotation of the reversal, $\tilde{y}_n$, having the smallest lexicographic value.

Finding this hinges on a few important facts about Christoffel words: The first is that a lower Christoffel word is smaller (in lexicographic order) than all of its non-trivial rotations (i.e. it is a Lyndon word). The second is that reversing a lower Christoffel word is equivalent to a rotation, i.e.

the sets of rotations of y_n and $\tilde{y}_n$ coincide. The third is that the reversal of a lower Christoffel word is an upper Christoffel word with the same slope, and an upper Christoffel word has a lexicographic value higher than all of its non-trivial rotations.

Therefore, if y_n is a lower Christoffel word, then $\tilde{y}_n$ is an upper Christoffel word, and since their sets of rotations coincide, the lexicographically minimal rotation of $\tilde{y}_n$ is simply y_n itself, because y_n is minimal among all its rotations, which includes $\tilde{y}_n$. Put simply, we have

$$\min_{1\leq k\leq n} r_k(\tilde{y}_n) = y_n,$$

where the minimum here is taken in lexicographic order, which exactly coincides with the order of the words interpreted as binary integers. Rewriting this in terms of integer representations gives

$$\min_{1\leq k\leq n} \sum_{i=1}^{n} 2^{i-1} Y_{k+i} = \sum_{i=1}^{n} 2^{n-i} Y_i$$

and the result follows. □

This allows us to formulate the main result of this section, which states that all maximally exploitative cycles are Christoffel words.

Theorem 6.1. *Given $n = a + b$ with $\gcd(a, b) = 1$, the maximally exploitative cycle (i.e. the cycle y_n having the largest cyclic minimum $\tau_{\min}(y_n)$ among all cycles with the same length n and density $\rho(y_n) = \frac{b}{a+b}$) is the Christoffel word $(c(a, b))$.*

Proof. A word y is *balanced* if, for any two factors of the same length, the number of 1s differs by at most 1. A standard result in the combinatorics of words is that all Christoffel words are balanced, and among all words with a fixed number of 1s and 0s (i.e. with the same length and density), the Christoffel word is the unique balanced word, up to rotation.

Let $y = c(a, b)$ and suppose that w is any word with the same length and density. If w is a rotation of y, then $\tau_{\min}(w) = \tau_{\min}(y)$, and we are done. Assume that w is not a rotation of y. Then w is not balanced. Denote the lexicographically minimal rotation of w as $r = \min_{1\leq k\leq n} r_k(w)$. As mentioned in the proof of Lemma 6.1, a word which is lexicographically smaller than all its non-trivial rotations is called a Lyndon word. A theorem of Borel and Laubie [29, 30] is that a Christoffel word is lexicographically maximal among all Lyndon words with the same length and density. Since r is a Lyndon word, $r < y$ and therefore $\tau_{\min}(w) < \tau_{\min}(y)$. □

These results give powerful tools for choosing trust thresholds based on maximum acceptable densities of defections. However, they also show that for any set of trust thresholds with $\tau_{\alpha_2} < \frac{1}{2}$, there exists a maximally exploitative cycle which can be sustained indefinitely. Moreover, if the game starts with m_0^Y above the limit cycle, then the limit cycle will always be approached from above, and so Y will never break trust if it plays a maximally exploitative cycle with $\tau_{\min}(y_n) \geq \tau_{\alpha_2}$.

Remark 6.1. *Theorem 6.1 establishes the limits of sustainable exploitation, creating a 'safe region' for exploitative strategies. This raises the question of whether such strategies can coexist with trust-based strategies in a well-mixed population, a question which we will return to in Section 8.*

7 Trust thresholds in noisy games

Section 5 establishes the cost of breaking and rebuilding trust through runs of consecutive defections and cooperations, and Section 6 tells us how a clever opponent can exploit a 4H strategy, potentially

without ever breaking trust. This begs the question: Why should a 4H strategy allow for any defection at all? Setting $\tau_{\alpha_2} < \frac{1}{2}$ always opens the possibility for Y to defect without breaking trust over a long enough game. One compelling reason relates to noise: In real life, even very cooperative relationships can be subject to mistakes and miscommunications. Cooperative relationships must be robust enough to tolerate these errors up to a certain point, or they become impossible to maintain in the long run.

Many strategies which excel in normal IPD tournaments perform quite poorly in IPD games where player decisions are subject to noise. For example, the normally dominant tit-for-tat strategy can be locked into an endless cycle of defection when playing against another tit-for-tat strategy, if random noise causes one of them to defect.

With this motivation, we consider noise as the primary justification for having a 4H strategy tolerate any defection in the first place. As such, it is useful to have some results which tell us how trust thresholds should be set given some assumption about the opponent's noise probability. For simplicity, we will assume that only Y is affected by noise, and suppose that its strategy is independent of X, which acts deterministically as a 4H strategy. We will assume that Y executes its intended move with probability q, and executes the opposite of its intended move with probability $1 - q$. To begin with, we will assume that Y intends to always cooperate, so all defections are noise-induced.

So, the move Y_i is an i.i.d. random variable following a Bernoulli distribution with probability q. Accordingly, the move sequence $y = Y_1 Y_2 \dots$ is a Bernoulli process. Consider the stationary distribution of the memory value m_n^Y as $n \to \infty$. In this case, the contribution of the initial memory value m_0 vanishes, and the stationary distribution of m^Y is the distribution of

$$m^Y = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{2^i} Y_{n-i}.$$

This is a (biased) Bernoulli convolution[3]. The following properties draw from classical results on Bernoulli convolutions established in the 1930s by Kershner and Wintner [31] and Erdős [32]. By linearity of expectation and independence, we have

$$E[m^Y] = \sum_{i=1}^{\infty} \frac{q}{2^i} = q,$$
$$\mathrm{Var}(m^Y) = \sum_{i=1}^{\infty} \frac{q(1-q)}{4^i} = \frac{q(1-q)}{3}.$$

In the special case that $q = \frac{1}{2}$, each binary digit is equally likely to be 0 or 1, so the distribution is uniform on $[0, 1]$, and the CDF $F_q(\tau) := \Pr(m^Y \leq \tau)$ is given by $F_{\frac{1}{2}}(\tau) = \tau$, where $\tau \in [0, 1]$.

When $q \neq \frac{1}{2}$, the distribution is singular continuous [31, 32] and has no density function in the usual sense, but its CDF is well-defined and can be computed through the recursion

$$F_q(\tau) = (1-q)F_q(2\tau) + qF_q(2\tau - 1),$$

with the boundary conditions $F_q(x) = 0$ for $x < 0$, and $F_q(x) = 1$ for $x > 1$. The recursive formula for the CDF has a simple interpretation. To simplify notation, let Y_{-1} denote the first bit of m^Y

[3]Note that in our application, the memory discounting factor is fixed at $\frac{1}{2}$, and q biases the resulting Bernoulli convolution. Many of the more interesting and surprising properties of Bernoulli convolutions are for the unbiased case, where $q = \frac{1}{2}$, and where the discounting factor takes other values in $(0, 1)$.

(i.e. the bit closest to the radix), which is 1 with probability q and 0 with probability $1-q$. Denote the left-shift operator $T : [0,1] \to [0,1]$, defined $T(0.Y_{-1}Y_{-2}\dots) = Y_{-2}Y_{-3}\dots$. If $Y_{-1} = 0$, then $T(m^Y) = 2m^Y$, and if $Y_{-1} = 1$, then $T(m^Y) = 2m^Y - 1$. Since m^Y has the same distribution as $T(m^Y)$, they have the same CDF, and the recursion follows.

To make the following analysis more tractable, we will assume that $\theta_2 = \theta_3 = 0$, so the trust breaking threshold is simply α_2, and the forgiveness threshold is β_3. The conclusions of the following analysis also hold in the more general case, because this represents a kind of lower bound (because if $\theta_3 > 0$, forgiveness becomes even more difficult). Assuming that the thresholds of X obey the no shortcut assumptions **(NS1)**–**(NS4)**, the values of α_2 and β_3 determines one of three general behavioral regimes:

1. If $\alpha_2 > \frac{1}{2}$, even a single defection will break trust. Forgiveness can be made arbitrarily expensive for Y depending on the value of β_3.

2. If $\frac{1}{4} < \alpha_2 \leq \frac{1}{2}$, two consecutive defections will break trust. The value of τ_{α_2} additionally determines a minimum number of cooperations which must occur between two defections to avoid breaking trust. We will denote this minimum number of consecutive defections k.

3. If $\alpha_2 \leq \frac{1}{4}$, then at least two consecutive defections are permitted before trust breaks (separated by a run of cooperations of a given minimum length).

Regime 1 is not relevant in noisy games because it has no tolerance to noise whatsoever. Regime 3 could of course be further subdivided into smaller reciprocal powers of 2, each of which allows for an additional consecutive defection. Lowering α_2 makes X more tolerant of noise-induced defections, but also makes it more vulnerable to exploitation. To compensate for this, X must raise the forgiveness threshold τ_{β_2} in order to punish trust-breaking in accordance with Corollary 5.3. However, this also has consequences for a cooperative Y. Although X may tolerate a higher level of noise, if Y is unlucky and has too many noise-induced defections in a row, the cost of rebuilding trust may be very high, and noise throughout the rebuilding leg can make it even longer and more costly. So, both sides may benefit from a compromise where X is relatively strict with its trust-breaking threshold α_2, but is also relatively lenient with its forgiveness threshold β_3. For this reason, regime 2 is of the greatest interest to us, because it offers the best possibility of compromise between tolerance of noise and vulnerability to exploitation.

7.1 The cost for a cooperative noisy opponent

Throughout this section we assume that α_2 is a dyadic rational of the form

$$\alpha_2 = \frac{2^k - 1}{2^{k+1}} = 0.0\underbrace{11\dots11}_{k \text{ times}}0\dots,$$

and β_3 is a dyadic rational of the form

$$\beta_3 = \frac{2^j - 1}{2^j} = 0.\underbrace{11\dots11}_{j \text{ times}}0\dots.$$

In other words, k or fewer cooperations between defections will break trust, and j consecutive cooperations are needed to rebuild trust after it is broken.

We are interested in finding the expected difference in payout per round, $E[\Delta\pi]$ (in a slight abuse of notation, in this section we take $\Delta\pi$ as the per-round payoff difference).

If Y always intends to cooperate, and has a noise rate of $1-q$, then the trust-breaking rate per round, λ_{break}, is bounded above by

$$\lambda_{\text{break}} \leq (1-q)F_q(2\alpha_2),$$

where $F_q(2\alpha_2)$ can be interpreted as the probability that a single defection will push m^Y below α_2, and $(1-q)$ is the probability of that defection occurring.

Remark 7.1. *By the same reasoning, the exact break rate is given by*

$$\lambda_{\text{break}} = (1-q)\Pr(m^Y < 2\alpha_2 \mid s_2 = 1).$$

In other words, the probability of breaking trust is the probability of a defection from a state where a single defection breaks the trust threshold, and where X is trusting, i.e. $s_2 = 1$. Finding this quasi-stationary distribution requires solving a boundary value problem for the singular Bernoulli convolution and has no closed form. We therefore use the unconditional probability $F_q(2\alpha_2) = 1-q^k$ as a rigorous upper bound.

Then $F_q(2\alpha_2) = (1-q)\sum_{i=0}^{k-1} q^i = 1-q^k$. This can easily be seen by the fact that $m^Y < 2\alpha_2$ if the first digit of m^Y is 0; otherwise if the first digit is 1, the second digit must be 0; otherwise if the first two digits are 1, the third digit must be 0, and so on. Therefore

$$\lambda_{\text{break}} = (1-q)(1-q^k).$$

To compute the forgiveness rate, we need the probability of j consecutive cooperations following a defection, which is given by

$$\lambda_{\text{forgive}} = q^j(1-q).$$

Using this, we can compute the expected time from trust-breaking to forgiveness. Since λ_{forgive} represents the stationary rate of entries into the forgiven state per round, its reciprocal is the full cycle length between forgiveness events. Then

$$\frac{1}{\lambda_{\text{forgive}}} = N_{\text{forgive}} + \frac{1}{1-q},$$

where N_{forgive} is the expected number of rounds from breaking to forgiveness, and $\frac{1}{1-q}$ is the expected waiting time for a defection. Solving for N_{forgive}, we obtain

$$N_{\text{forgive}} = \frac{1}{q^j(1-q)} - \frac{1}{1-q} = \frac{1-q^j}{q^j(1-q)}.$$

Then we have

$$\Pr_{\text{distrust}} = \frac{\lambda_{\text{break}} N_{\text{forgive}}}{1+\lambda_{\text{break}} N_{\text{forgive}}} \leq \frac{(1-q^k)(1-q^j)}{q^j + (1-q^k)(1-q^j)},$$
$$\Pr_{\text{trust}} = 1 - \Pr_{\text{distrust}} \geq \frac{q^j}{q^j + (1-q^k)(1-q^j)},$$

where $\Pr_{\text{distrust}}$ is the stationary probability that X is in the distrusting state, i.e. that X defects while Y cooperates with probability q. Similarly, $\Pr_{\text{trust}}$ is the stationary probability that X is in the trusting state. Then the stationary round outcome probabilities are given by

$$\Pr(CD) = (1-q)\Pr_{\text{trust}}, \quad \Pr(DC) = q\Pr_{\text{distrust}}.$$

Using this, we can compute the stationary expected payoff difference per round, $E[\Delta\pi]$:

$$E[\Delta\pi] = (T-S)(q \Pr_{\text{distrust}} -(1-q) \Pr_{\text{trust}}) = (T-S)(\Pr_{\text{distrust}} -(1-q)).$$

We will define the *strictness index* Σ as

$$\Sigma := \frac{E[\Delta\pi]}{T-S} = \Pr_{\text{distrust}} -(1-q),$$

and we will say that X is *strict* if $\Sigma > 0$, *lenient* if $\Sigma < 0$, and *fair* if $\Sigma = 0$ (or if $\Sigma \approx 0$.) This has a very transparent behavioral interpretation. If $\Sigma > 0$, then X has a higher long-run payoff because its trust and forgiveness thresholds are such that it spends more rounds punishing Y than Y spends betraying X; if $\Sigma < 0$, then X spends fewer rounds punishing Y than Y spends betraying X; and if $\Sigma = 0$, then X punishes Y as much as Y betrays X.

Fairness clearly requires $\Pr_{\text{distrust}} = (1-q)$. Substitution gives the upper bound

$$\frac{E[\Delta\pi]}{T-S} \leq \frac{(1-q^k)(1-q^j)}{q^j + (1-q^k)(1-q^j)} - (1-q).$$

This lets us prove the following proposition:

Proposition 7.1. *Given $\alpha_2 = \frac{2^k-1}{2^{k+1}}$ and $\beta_3 = \frac{2^j-1}{2^j}$, and assuming that $\theta_2 = \theta_3 = 0$ and that X's thresholds satisfy **(NS1)**—**(NS4)**, then*

$$\frac{(1-q^{k-1})(1-q^j)}{q^j + (1-q^{k-1})(1-q^j)} - (1-q) \leq \Sigma \leq \frac{(1-q^k)(1-q^j)}{q^j + (1-q^k)(1-q^j)} - (1-q).$$

Proof. We have already established the upper bound. All that remains is to show that the lower bound holds. For this, we return to the exact break rate as given in Remark 7.1, and note that the trusting state implies that $m^Y \geq \alpha_2$. Therefore

$$\lambda_{\text{break}} = (1-q)\Pr(m^Y < 2\alpha_2 \mid s_2 = 1) \geq (1-q)\Pr(m^Y < 2\alpha_2 \mid m^Y \geq \alpha_2).$$

The probability $\Pr(m^Y < 2\alpha_2 \mid m^Y \geq \alpha_2)$ can be computed as

$$\Pr(m^Y < 2\alpha_2 \mid m^Y \geq \alpha_2) = \frac{F_q(2\alpha_2) - F_q(\alpha_2)}{1 - F_q(\alpha_2)} = \frac{q(1-q^k)}{1-(1-q)(1-q^k)}.$$

We claim that

$$\frac{q(1-q^k)}{1-(1-q)(1-q^k)} \geq 1 - q^{k-1}.$$

Indeed, cross-multiplying (since both sides are positive), we obtain

$$q(1-q^k) \geq (1-q^{k-1})\big(1-(1-q)(1-q^k)\big).$$

Expanding the right-hand side:

$$(1-q^{k-1})\big(1-(1-q)(1-q^k)\big) = q - q^k + (1-q)q^k - (1-q)q^{2k-1} = q - q^{k+1} - (1-q)q^{2k-1}.$$

So the inequality reduces to

$$q - q^{k+1} \geq q - q^{k+1} - (1-q)q^{2k-1},$$

which is equivalent to

$$0 \geq -(1-q)q^{2k-1},$$

which holds because the right-hand side is negative. So the inequality is strict for all $q \in (0,1)$. Therefore,

$$(1-q)(1-q^{k-1}) \leq \lambda_{\text{break}} \leq (1-q)(1-q^k),$$

and the result follows from the monotonicity of $E[\Delta\pi]$ with respect to λ_{break}. □

This result has a number of interesting consequences and implications. First of all, it implies that a given level of strictness can (approximately) be attained by different values of j and k. This corresponds with our earlier discussion, in the sense that low k and high j correspond to α_2 and β_3 which are far apart, meaning that trust is broken less frequently, but the consequences for breaking it are more severe (this also means that longer games are required to approach the limit $E[\Delta\pi] = (T-S)\Sigma$.) The value of j corresponding to $k=1$ for a given fairness is the widest spread that the two thresholds can have, although we note that this result only applies to the regime where $\frac{1}{4} < \alpha_2 \leq \frac{1}{2}$. Otherwise, if more than one consecutive defection is allowed, the assumption on the dyadic form of α_2 must be adjusted accordingly (see Remark 7.2 for a brief discussion.)

At the other extreme, consider the limit of the upper and lower bounds as $k \to \infty$ (corresponding to $\alpha_2 \to \frac{1}{2}$). In this case, by the squeeze theorem, Σ converges to

$$\Sigma \to \frac{1-q^j}{q^j + (1-q^j)} - (1-q) = q - q^j.$$

This has two interesting implications. Firstly, as k increases, the distance between the upper and lower bounds of Σ decays geometrically, so these thresholds become more reliable bounds on Σ. Secondly, fairness in the $k \to \infty$ limit is possible only when $j=1$, which implies that $\alpha_2 = \beta_3 = \frac{1}{2}$. In this case, there is no hysteresis, and X is exactly the classical tit-for-tat strategy. Accordingly, fair (or approximately fair) 4H strategies can be thought of as a kind of hysteretic " widening" of tit-for-tat which is tolerant of some degree of noise from Y.

Remark 7.2. *In order to consider the regime where $\alpha_2 \leq \frac{1}{4}$, we may take α_2 to be of the dyadic form*

$$\alpha_2 = \frac{2^k - 1}{2^{k+\ell}} = 0.\ \underbrace{0\ldots0}_{\ell \text{ times}}\ \underbrace{1\ldots1}_{k \text{ times}}\ 0.$$

In this case, $F_q(\alpha_2) = (1-q)^\ell F_q(2^\ell \alpha_2) = (1-q)^\ell(1-q^k)$, and so

$$\lambda_{\text{break}} \leq (1-q)^\ell(1-q^k).$$

Following the same reasoning as earlier in the section, we can obtain the upper bound

$$\Sigma \leq \frac{(1-q^k)(1-q^j)(1-q)^\ell}{q^j + (1-q^k)(1-q^j)(1-q)^\ell} - (1-q).$$

However, obtaining a corresponding lower bound is not so simple, as the method used in the proof of Proposition 7.1 implicitly assumed $\ell = 1$.

Remark 7.3. *Proposition 7.1 also has a clear evolutionary interpretation. A fair strategy ($\Sigma = 0$) neither profits from nor subsidizes its partner's noise, and may therefore be more attractive as a long-term partner when partners can choose whom to associate with. A strict strategy ($\Sigma > 0$) profits from Y's noise, creating an incentive for Y to reduce its noise rate—or to seek a more lenient partner. A lenient strategy ($\Sigma < 0$) subsidizes Y's defection, creating a moral hazard. These interpretations are conjectural; we return to them in the discussion.*

8 Conclusions and discussion

In this paper, we introduced a novel family of trust-based strategies in the iterated prisoner's dilemma, and developed a geometric framework for their analysis. The defining properties of this family—internal state, path dependence, and asymmetry—are drawn from existing characterizations of trust (particularly partner-specific dyadic trust, as opposed to public reputation) in psychology, behavioral economics, and evolutionary game theory, and align closely with the mathematical notion of hysteresis. We realized the properties concretely through the hysteron—the elementary non-ideal relay—and studied the resulting four hysteron (4H) strategy in detail.

The central technical innovation is the memory square: a geometric representation in which game histories are points, thresholds are lines, and round outcomes are affine contractions. This representation transforms the study of a long-memory strategy into a tractable geometric problem, and allows us to prove rigorous results that would be difficult or impossible to obtain for other long-memory strategies such as finite-state machines or artificial neural networks. The tractability of the analysis across different timescales is, we believe, the distinguishing feature of this strategy family.

We analyzed the 4H strategy from three complementary perspectives. First, we proved a deterrence theorem, establishing conditions under which betrayal is unprofitable for the exploiter. This formalizes the intuitive idea that trust is more expensive to rebuild than it is profitable to break, and gives explicit parameter conditions for this asymmetry. Second, we characterized the maximally exploitative cycles that an opponent can sustain indefinitely without breaking trust, showing that these are exactly the lower Christoffel words. This reveals a deep connection between the IPD and the combinatorics of words, and establishes the limits of sustainable exploitation. Third, we analyzed noise tolerance, deriving conditions on the thresholds that balance resilience to noise-induced defections against vulnerability to intentional exploitation. We defined a strictness index and showed that a fair, strict, or lenient posture can be achieved at different levels of noise tolerance, where fairness with zero noise tolerance is connected to the classical tit-for-tat strategy as a limiting case.

8.1 Evolutionary interpretation

These three results each have direct evolutionary interpretations, even though we have not performed evolutionary simulations in this paper. This is because the payoff of an agent and the payoff difference between two agents directly correlate with evolutionary fitness. The deterrence theorem identifies payoff asymmetries which would enter a replicator or invasion analysis. The Christoffel word characterization of optimal patterned exploitation establishes a safe region for exploitative strategies, raising the question of whether such strategies can coexist with trust-based strategies. The fairness analysis suggests two distinct mechanisms by which fairness could evolve. First, lenience creates a moral hazard [33]: if X is lenient, Y has no incentive to reduce its noise rate (and has incentives to intentionally defect), so lenient strategies are costly in the long tun. This eliminates lenience, but does not by itself favor fairness over strictness. Second, in games with partner choice[15, 16, 17], fairness may be more attractive as a long-term posture than either lenience (which is exploited) or strictness (which extracts a surplus). Fairness is therefore the intermediate posture that avoids both the moral hazard of lenience and the surplus extraction of strictness. We do not test these conjectures here, but they suggest natural directions for future work.

8.2 Future work

In addition to the above conjectures, several interesting questions remain open. We have not studied the evolutionary stability of 4H strategies against diverse populations, nor their performance in genetic algorithm tournaments. The noise analysis assumes an always-cooperating opponent with a fixed noise rate; a natural extension is to consider a general memory-one opponent, which would require computing the quasi-stationary distribution of the 4H strategy viewed as a piecewise-deterministic Markov process [34, 35]. Finally, the hysteron outputs could be inverted to create exploitative strategies that defect against cooperating opponents until punished, and these inverted strategies, together with the original 4H strategies, form a broader family that remains tractable within the geometric framework described here. We hope that this paper provides both a useful set of tools and a clear agenda for future work on trust-based strategies in iterated games.

References

[1] Robert Axelrod. Effective choice in the prisoner's dilemma. *Journal of Conflict Resolution*, 24(1):3–25, 1980. doi: 10.1177/002200278002400101.

[2] Robert Axelrod. *The Evolution of Cooperation*. Basic Books, 1984.

[3] Robert Axelrod and Douglas Dion. The further evolution of cooperation. *Science*, 242(4884): 1385–1390, 1988. doi: 10.1126/science.242.4884.1385.

[4] Martin Nowak and Karl Sigmund. A strategy of win-stay, lose-shift that outperforms tit-for-tat in the prisoner's dilemma game. *Nature*, 364:56–58, 1993. doi: 10.1038/364056a0.

[5] William H. Press and Freeman J. Dyson. Iterated prisoner's dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109 (26):10409–10413, 2012. doi: 10.1073/pnas.1206569109.

[6] Martin A. Nowak and Karl Sigmund. Evolution of indirect reciprocity. *Nature*, 437:1291–1298, 2005. doi: 10.1038/nature04131.

[7] Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006. doi: 10.1126/science.1133755.

[8] Pedro Dal Bó and Guillaume R. Fréchette. On the determinants of cooperation in infinitely repeated games: A survey. *Journal of Economic Literature*, 56(1):60–114, 2018. doi: 10.1257/jel.20160980.

[9] Robert Hoffmann. Twenty years on: The evolution of cooperation revisited. *Journal of Artificial Societies and Social Simulation*, 3(2):1–13, 2000. URL `https://www.jasss.org/3/2/forum/1.html`.

[10] Denise M. Rousseau, Sim B. Sitkin, Ronald S. Burt, and Colin Camerer. Not so different after all: A cross-discipline view of trust. *Academy of Management Review*, 23(3):393–404, 1998. doi: 10.5465/amr.1998.926617.

[11] Roger C. Mayer, James H. Davis, and F. David Schoorman. An integrative model of organizational trust. *Academy of Management Review*, 20(3):709–734, 1995. doi: 10.5465/amr.1995.9508080335.

[12] Joyce Berg, John Dickhaut, and Kevin McCabe. Trust, reciprocity, and social history. *Games and Economic Behavior*, 10(1):122–142, 1995. doi: 10.1006/game.1995.1027.

[13] Caitlin Duncan, Lorena Sganzerla, Laura Kaltwasser, and Isabel Dziobek. Trust learning in the repeated trust game: A meta-analytic study. *British Journal of Psychology*, 117(3):1026–1055, 2025. doi: 10.1111/bjop.70045.

[14] Dongqing Li. Emergence of cooperation in trust-driven coevolutionary networks. *Chaos, Solitons & Fractals*, 208:118274, 2026. doi: 10.1016/j.chaos.2026.118274.

[15] James J. Bull and William R. Rice. Distinguishing mechanisms for the evolution of cooperation. *Journal of Theoretical Biology*, 149(1):63–74, 1991. doi: 10.1016/S0022-5193(05)80072-4.

[16] Ronald Noë and Peter Hammerstein. Biological markets: Supply and demand determine the effect of partner choice in cooperation, mutualism and mating. *Behavioral Ecology and Sociobiology*, 35:1–11, 1994. doi: 10.1007/BF00167053.

[17] Pat Barclay. Competitive helping increases with the size of biological markets and invades defection. *Journal of Theoretical Biology*, 281(1):47–55, 2011. doi: 10.1016/j.jtbi.2011.04.023.

[18] Tadeas Priklopil, Krishnendu Chatterjee, and Martin A. Nowak. Optional interactions and suspicious behaviour facilitates trustful cooperation in prisoner's dilemma. *Journal of Theoretical Biology*, 433:64–72, 2017. doi: 10.1016/j.jtbi.2017.08.025.

[19] Yohsuke Murase. Costly punishment sustains indirect reciprocity under low defection detectability. *Journal of Theoretical Biology*, 600:12043, 2025. doi: 10.1016/j.jtbi.2025.112043.

[20] Zhao Song, Chen Shen, and The Anh Han. Network reciprocity turns cheap talk into a force for cooperation. *Journal of Theoretical Biology*, 617:112303, 2026. doi: 10.1016/j.jtbi.2025.112303.

[21] David P. Kraines and Vivian Y. Kraines. Natural selection of memory-one strategies for the iterated prisoner's dilemma. *Journal of Theoretical Biology*, 203(4):335–355, 2000. doi: 10.1006/jtbi.2000.1089.

[22] Marko Jurisic, Dragutin Kermek, and Mladen Konecki. A review of iterated prisoner's dilemma strategies. In *MIPRO 2012 - 35th International Convention*, pages 1093–1097. IEEE, 2012.

[23] James C. Cox. How to identify trust and reciprocity. *Games and Economic Behavior*, 46(2):260–281, 2004. doi: 10.1016/S0899-8256(03)00119-2.

[24] Mark A. Krasnosel'skii and Alexei V. Pokrovskii. *Systems with Hysteresis*. Springer, 1989. doi: 10.1007/978-3-642-61302-9.

[25] Augusto Visintin. *Differential Models of Hysteresis*. Springer, 1994. doi: 10.1007/978-3-662-11557-2.

[26] Isaak D. Mayergoyz. *Mathematical Models of Hysteresis and Their Applications*. Academic Press, 2003. doi: 10.1016/B978-0-12-480873-7.X5000-2.

[27] M. Lothaire. *Algebraic Combinatorics on Words*. Cambridge University Press, 2002. doi: 10.1017/CBO9781107326019.

[28] Jean Berstel, Aaron Lauve, Christophe Reutenauer, and Franco V. Saliola. *Combinatorics on Words: Christoffel Words and Repetitions in Words*. American Mathematical Society, 2008. doi: 10.1090/crmm/027.

[29] Christophe Reutenauer. *From Christoffel Words to Markoff Numbers*. Oxford University Press, 2019. doi: 10.1093/oso/9780198827542.001.0001.

[30] J.-P. Borel and F. Laubie. Quelques mots sur la droite projective réelle. *Journal de théorie des nombres de Bordeaux*, 5(1):23–51, 1993. doi: 10.5802/jtnb.77.

[31] Richard Kershner and Aurel Wintner. On symmetric Bernoulli convolutions. *American Journal of Mathematics*, 57(3):541–548, 1935. doi: 10.2307/2371185.

[32] Paul Erdős. On a family of symmetric Bernoulli convolutions. *American Journal of Mathematics*, 61(4):974–976, 1939. doi: 10.2307/2371641.

[33] Bengt Holmström. Moral hazard and observability. *The Bell Journal of Economics*, 10(1): 74–91, 1979.

[34] Pierre Collet, Servet Martínez, and Jaime San Martín. *Quasi-Stationary Distributions: Markov Chains, Diffusions and Dynamical Systems*. Springer, 2012. doi: 10.1007/978-3-642-33131-2.

[35] Mark H. A. Davis. *Markov Models and Optimization*. Chapman and Hall, 1993. doi: 10.1007/978-1-4899-4483-2.